

AlphaStack: AI-powered project generator

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1 Abstract

This paper presents AlphaStack, a novel approach to autonomous code generation using multi-agent systems with iterative self-healing and comprehensive validation across diverse programming paradigms. It transforms natural language descriptions into complete, production-ready codebases with Docker configurations and automated testing.

2 Introduction

Modern software development involves boilerplate code, setup, and configuration. We introduce AlphaStack, a multi-agent framework that uses planning and correction agents. This autonomous multi-agent framework addresses key challenges in AI-assisted software development through separation of planning and correction concerns.

3 Methodology

The AlphaStack pipeline involves several phases. First, Blueprint Generation analyzes requirements and creates a software architecture. Next, it generates multi-file code configurations. Docker integration allows for automated environment creation and isolated build testing.

4 Architecture Diagram

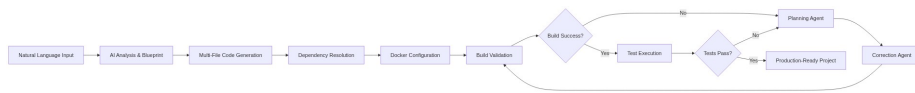


Figure 1: AlphaStack Architecture Diagram

5 Results

We evaluated our methodology on HumanEval and MDDP datasets across various modern language models: GPT-5.2, GLM-5, MiniMax-m2.5, and Claude Sonnet 4.6.

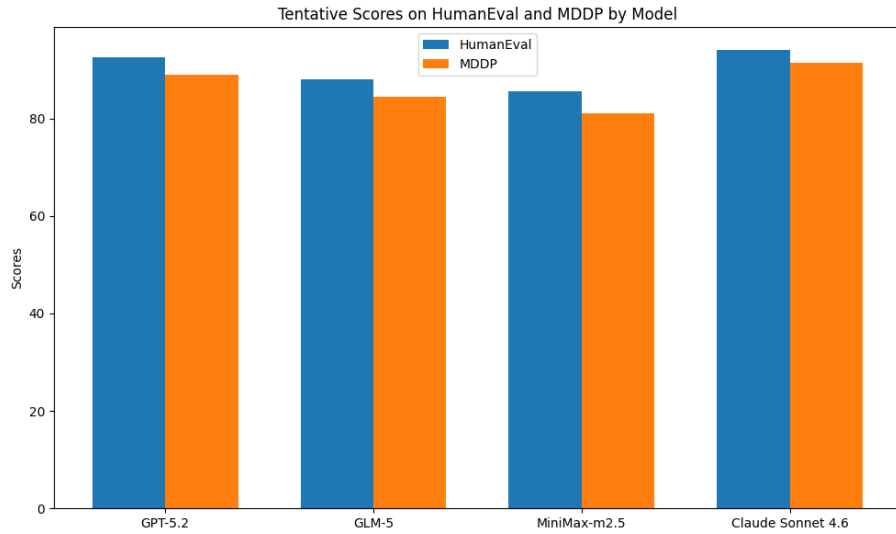


Figure 2: Tentative evaluation results on HumanEval and MDDP

6 Conclusion

AlphaStack successfully demonstrates iterative self-healing and autonomous code generation capabilities. By employing Docker-based validation and separated agent roles, the system achieves robust code generation in complex scenarios. Future work will expand programming language support and evaluation challenges.